AWS HANDS-ON LAB

Highly Available ECS Cluster

Application Load Balancer + Dynamic Port Mapping

Implementation guide • Error review • Validation checklist

LAB OUTCOME
The supplied lab record reports 6 ECS tasks running, 6 target-group targets healthy, and the Hiring application accessible through the ALB DNS name at /hiring/. This document organizes the configuration, validation points, and troubleshooting lessons into a reusable runbook.

At a glance

Item	Configured value
AWS Region	us-east-1 (US East — N. Virginia)
ECS launch type	EC2
Cluster / Service	hiring-app-cluster / hiring-app-service
Application image	sabair0509/hiring-app:works
Desired task count	6
Application container port	8080
Network mode / host port	bridge / 0 (dynamic host port)
Load balancer / target group	hiring-app-alb / hiring-app-tg
Application context path	/hiring/

Prepared as a practical implementation record

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1. Objective and success criteria

Deploy the container image sabair0509/hiring-app:works on Amazon ECS using the EC2 launch type, fronted by an internet-facing Application Load Balancer (ALB). Configure bridge networking and host port 0 so ECS can register each task against a dynamically allocated host port.

- Maintain a desired service count of six tasks, with the service scheduler replacing failed tasks.
- Distribute tasks across at least two Availability Zones where capacity and placement permit.
- Route public HTTP traffic through the ALB only to healthy registered targets.
- Validate the deployed application at the /hiring/ context path.

Important wording
A desired count of six is not an absolute guarantee that six tasks will remain running at every instant. ECS can temporarily run fewer tasks during failures, deployments, capacity shortages, or unhealthy replacements. Verify service events and Auto Scaling capacity when the count drops.

2. Purpose and architecture

The ALB provides a single public entry point and routes requests to healthy targets. ECS runs multiple copies of the container on EC2 container instances. With bridge mode and hostPort 0, the container still listens on port 8080, while Docker/ECS allocates a host-side ephemeral port for each task.

```
Client browser
    |
    | HTTP :80
    v
Application Load Balancer (hiring-app-alb)
    |
    | Target group: hiring-app-tg
    | Registered as EC2-instance-IP : dynamic-host-port
    v
ECS service (desired count = 6)
    |-------------------------------|
    v                               v
EC2 container instance A       EC2 container instance B
  Task -> host port :327xx        Task -> host port :32xxx
  Container :8080                 Container :8080
  Tomcat /hiring/                 Tomcat /hiring/
```

Traffic path to remember
Browser → ALB listener :80 → target group → EC2 instance at dynamically assigned host port → container port 8080 → Tomcat application context /hiring/.

High availability depends on using subnets in multiple AZs, healthy ALB targets in those AZs, adequate EC2 capacity, and a service placement strategy that spreads tasks. Confirm actual task placement in the ECS service and task details; selecting two subnets alone does not prove balanced placement.

3. Prerequisites and capacity planning

Prerequisite	Lab value / note
Region	us-east-1
VPC	vpc-0fff3eb0730a112f0
Availability Zones	us-east-1a and us-east-1b
EC2 instance type	t3.small; 2 vCPU and 2 GiB memory per instance
Cluster Auto Scaling group	Minimum 2, maximum 4 instances in the supplied configuration
Task definition memory	0.25 GB per task as configured in the lab
Image	sabair0509/hiring-app:works
IAM	ECS task execution / container-instance roles with required permissions
Logging	CloudWatch log group /ecs/hiring-app-task

Capacity note: six tasks × 0.25 GiB = 1.5 GiB of task memory reservations in aggregate. That arithmetic does not by itself prove the tasks fit on two 2-GiB instances: the operating system, ECS agent, Docker, other reservations, CPU, and placement constraints also consume capacity. Validate actual registered resources and service events. EC2, ALB, EBS, public IPv4, data transfer, and CloudWatch usage may incur charges; t3.small is not universally free-tier eligible.

4. Step-by-step implementation

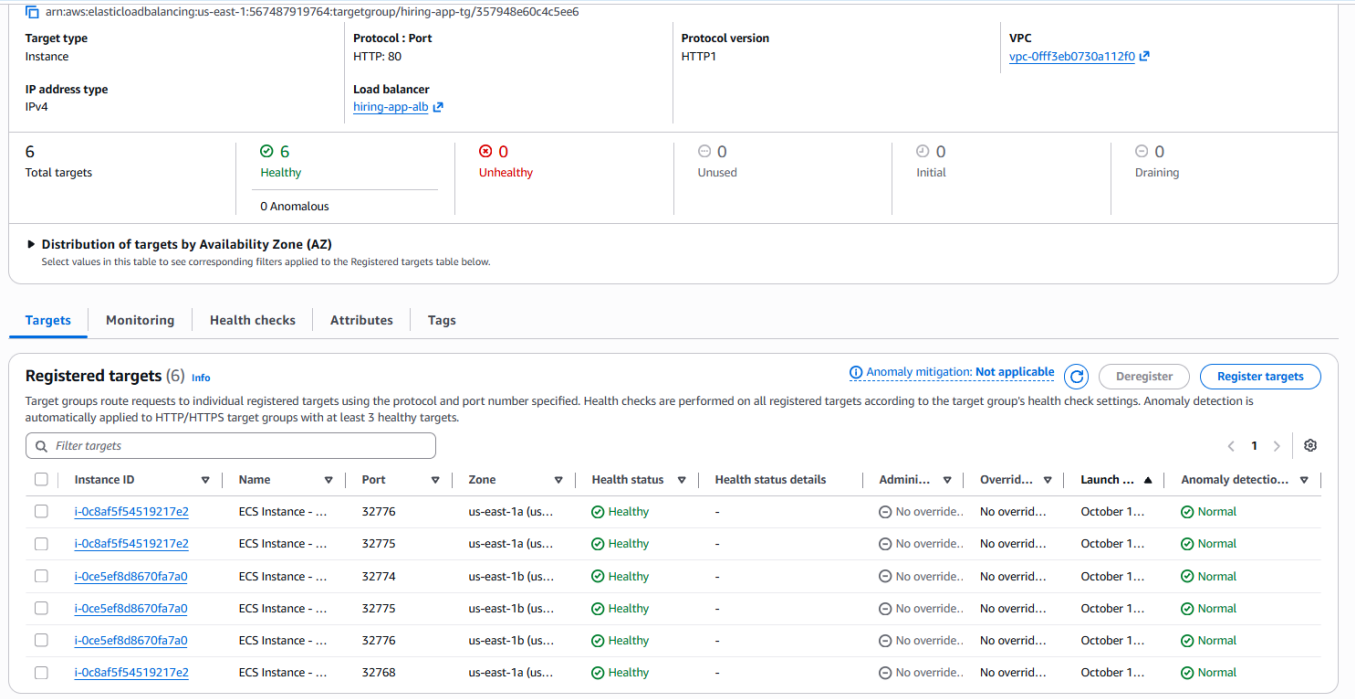
Step 1 — Create the ALB and target group

Console path: EC2 → Load Balancers → Create load balancer → Application Load Balancer

Setting	Value
Load balancer name	hiring-app-alb
Scheme	Internet-facing
VPC	vpc-0fff3eb0730a112f0
AZ/subnets	Subnets in us-east-1a and us-east-1b
ALB security group	hiring-app-alb-sg
Inbound rule	HTTP TCP 80 from 0.0.0.0/0 (restrict further for production as appropriate)

Setting	Value
Listener	HTTP :80
Target group	hiring-app-tg
Target type	Instances
Protocol / initial port	HTTP :80
Initial health path	/ (corrected in Step 5)

Screenshot evidence
EC2 → Target Groups → hiring-app-tg → Details: show Target type = Instances and the configured HTTP protocol/port. The target group's default port is not the same as the dynamically registered port for each ECS task.



Step 2 — Create the ECS cluster and EC2 capacity

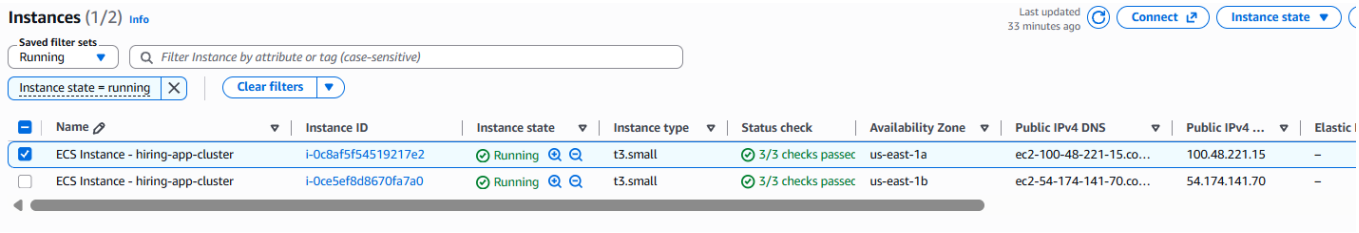
Console path: ECS → Clusters → Create cluster

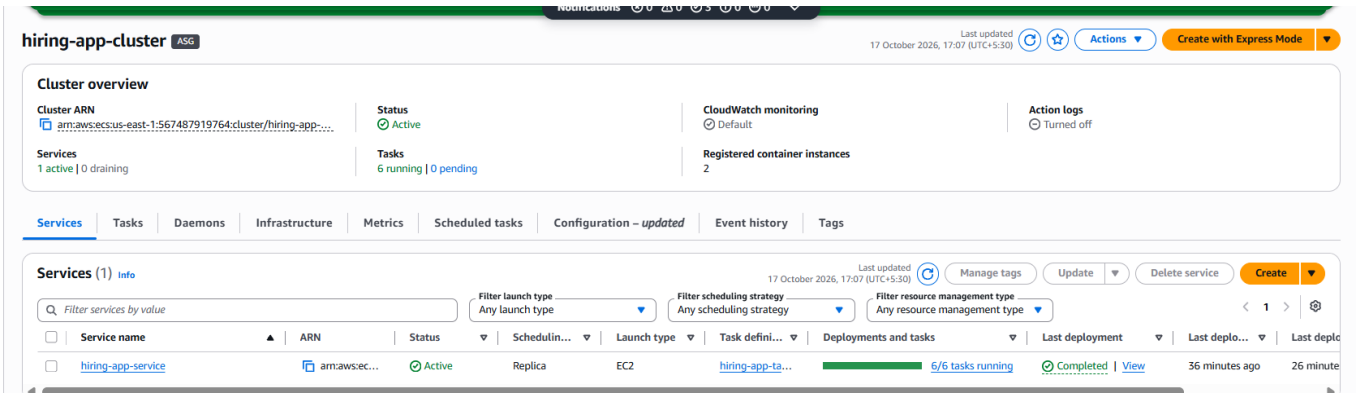
Setting	Value
Cluster name	hiring-app-cluster
Infrastructure	ECS on self-managed EC2 instances; console may also display Fargate as an available option
Auto Scaling group	Create new; minimum 2, maximum 4 (as supplied)
Instance type	t3.small
Instance role	ECS container-instance role / console-created default role
VPC and subnets	vpc-0fff3eb0730a112f0; us-east-1a and us-east-1b

Setting	Value
Instance security group	hiring-app-instance-sg
Inbound from ALB	Allow the application traffic range from source security group hiring-app-alb-sg

For dynamic port mapping, the instance security group must allow traffic from the ALB security group to the host ports used by ECS tasks. The lab used an All TCP rule from hiring-app-alb-sg. This is broad; a tighter production rule should allow only the necessary dynamic host-port range supported by the container-instance configuration and ECS networking design.

Screenshot evidence
EC2 → Instances: show the ECS container instances are running and their security groups include hiring-app-instance-sg. Also verify the cluster's registered container instances in ECS → Cluster → Infrastructure/Container instances.





Step 3 — Create the task definition

Console path: ECS → Task definitions → Create new task definition

Setting	Value
Family	hiring-app-task
Compute / launch compatibility	Amazon EC2
Network mode	bridge
Task CPU	0.25 vCPU (as supplied)
Task memory	0.25 GB (as supplied; confirm supported units and reservation in the revision)
Container name	hiring-app
Image URI	sabair0509/hiring-app:works
Container port	8080

Setting	Value
Host port	0
Log driver / group	awslogs /ecs/hiring-app-task

Why host port 0 matters
The application listens inside the container on port 8080. With bridge mode and hostPort 0, the runtime assigns an available host port and ECS registers the instance plus that port in the target group. Do not confuse this with awsvpc networking, where each task has its own network interface and port mapping behaves differently.

Inspect the task definition JSON. The important fields should be equivalent to:

```
"networkMode": "bridge",
"portMappings": [
  {
    "containerPort": 8080,
    "hostPort": 0,
    "protocol": "tcp"
  }
]
```

Screenshot evidence
ECS → Task definitions → hiring-app-task → selected revision → JSON: capture networkMode = bridge and the port mapping with containerPort 8080 / hostPort 0.

Step 4 — Create the ECS service

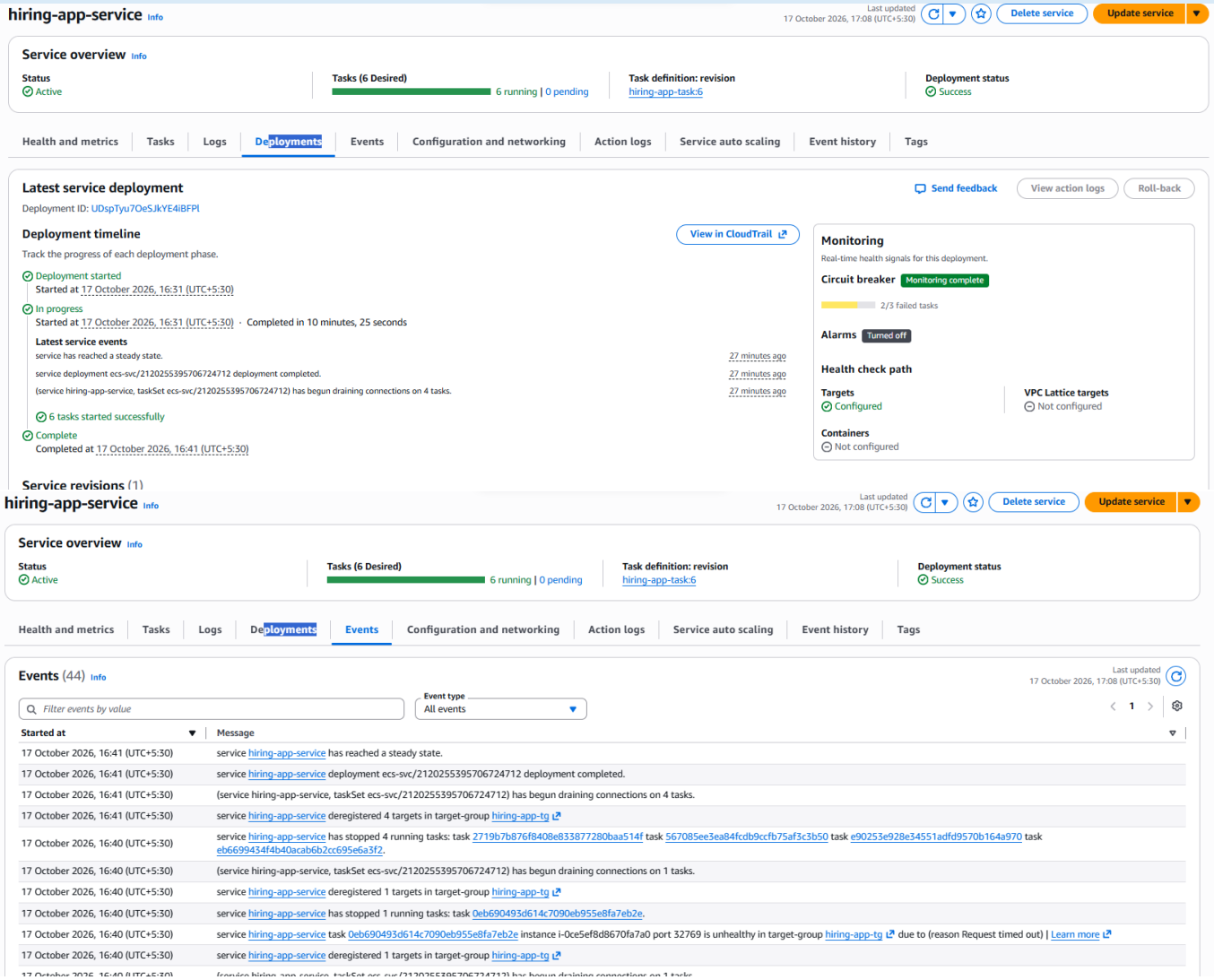
Console path: ECS → Clusters → hiring-app-cluster → Services → Create

Setting	Value
Service name	hiring-app-service
Desired tasks	6
Availability Zone rebalancing	Enabled, if available for the selected service configuration
Load balancing	Attach hiring-app-alb and target group hiring-app-tg
Container mapping	Select container hiring-app and container port 8080
Placement strategy	Spread across Availability Zones; add a suitable spread across instances if needed
Subnets	Subnets in us-east-1a and us-east-1b

When configuring the ALB integration, choose the container port 8080. ECS uses the task definition's port mapping and registers the dynamically assigned host port for each task. The console may summarize the container mapping differently across UI versions; inspect the task definition and registered target ports rather than relying on a label such as 8080:8080 alone.

Screenshot evidence

ECS → Cluster → Services → hiring-app-service: show desired/running task counts, load-balancer attachment, and placement settings. ECS → Tasks: show task AZs and the EC2 container instances on which tasks run.



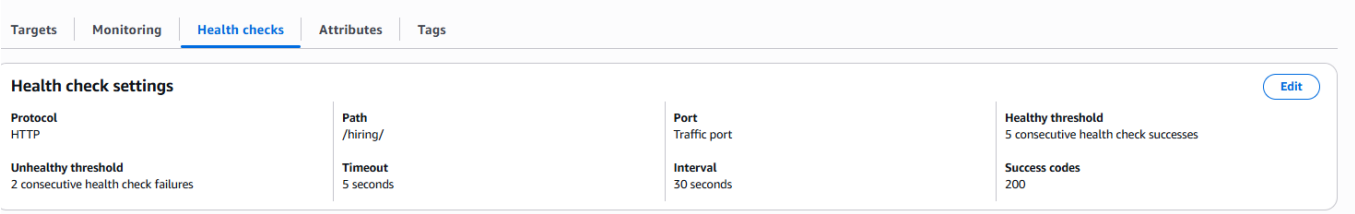
Step 5 — Correct the target-group health check

Console path: EC2 → Target Groups → hiring-app-tg → Health checks → Edit

Setting	Final value
Health check protocol	HTTP
Health check path	/hiring/
Success codes	200
Port	Traffic port / registered target port

The application WAR is named hiring.war. In a conventional Tomcat deployment, this normally creates the /hiring/ context path. A request to / may therefore return 404 even while Tomcat is running correctly. Set the health-check path to /hiring/ and confirm that the application returns an accepted success code.

Screenshot evidence
EC2 → Target Groups → hiring-app-tg → Health checks: show path /hiring/ and success code 200. Then open the Targets tab and capture all six targets with Healthy status and their registered high ports.



5. Five key mistakes and how to diagnose them

Use this table as an error-review checklist. Check the listed console location first, then use logs or commands to isolate the cause. The symptom can have more than one cause, so confirm before changing settings.

#	Mistake / misconfiguration	Typical symptom	Where to check / corrective action
1	Task memory reservation set too high (for example, 0.5 GB per task)	Service events report insufficient memory; tasks remain pending or fail placement.	ECS → Cluster → Service → Events; task definition revision; container-instance registered memory. Reduce reservation only if the application can safely run within it, or add capacity.
2	Configured container port as 80 instead of the app's 8080	Target checks fail; connection refused or timeout; app logs show Tomcat on 8080.	ECS → Task definition → container port mapping; CloudWatch log group /ecs/hiring-app-task; look for http-nio-8080. Set containerPort to 8080.
3	Instance security group blocks ALB access to dynamic host ports	Targets show Request timed out or fail health checks.	EC2 → Instances → select instance → Security tab → instance SG inbound rules. Permit traffic from hiring-app-alb-sg to the necessary host-port range.
4	Auto Scaling group is also registering instances directly into the target group	Unexpected port-80 instance targets appear and remain unhealthy alongside ECS task targets.	EC2 → Auto Scaling Groups → select ASG → Integrations / Load balancing. Remove the target-group association if ECS service-managed registration is intended.
5	Health check uses / rather than /hiring/	HTTP 404 / health check failure even though the container and Tomcat are running.	EC2 → Target Groups → hiring-app-tg → Health checks; test the app path directly. Set path to /hiring/ if that endpoint returns the configured success code.

Diagnostic sequence

1. Check ECS service Events for placement, image-pull, health-check, or capacity errors.
2. Open a stopped task and inspect its Stopped reason and container exit code.
3. Confirm the task definition revision, image tag, network mode, and port mapping.
4. Check CloudWatch logs for startup errors and the Tomcat listening port.
5. Inspect target-group target health reason codes and registered target ports.
6. Validate ALB and instance security-group rules, then test the application context path.

6. Testing and final verification

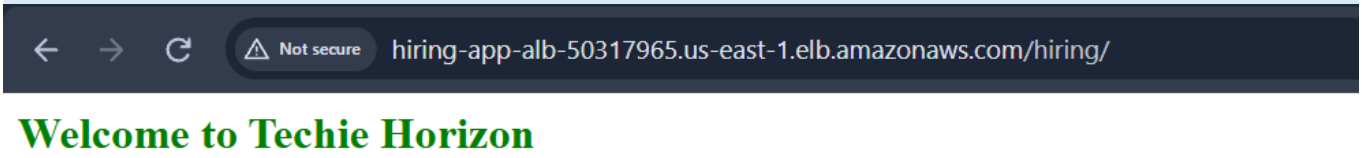
Check	Expected result	Console / test location
Service desired count	6	ECS → Cluster → Services → hiring-app-service
Running tasks	6 after service stabilizes	ECS → Service → Tasks
AZ distribution	Tasks placed across us-east-1a and us-east-1b, subject to capacity	ECS task details; container instance AZ
Target registration	Six task targets, each with a dynamic host port	EC2 → Target Groups → hiring-app-tg → Targets
Target health	6 Healthy	Target group → Targets / Health status
Application URL	ALB DNS name + /hiring/	Browser or curl from a permitted client
Application startup log	Tomcat connector starts on port 8080	CloudWatch → Log groups → /ecs/hiring-app-task

Application endpoint recorded in the supplied lab notes:

http://hiring-app-alb-50317965.us-east-1.elb.amazonaws.com/hiring/

If the endpoint changes or the ALB is recreated, copy the current DNS name from EC2 → Load Balancers rather than relying on the recorded URL.

Health status timing
After changing health-check settings or registering tasks, allow a few health-check intervals for the target group to converge. If targets remain unhealthy, use the target's reason code and last health-check response; do not assume waiting alone will fix a port, path, security-group, or application error.



7. Troubleshooting decision guide

Observed result	Likely area	Next action
Tasks stay Pending	Capacity, task size, placement constraints, instance registration	Read ECS service Events; compare requested task CPU/memory with available container-instance resources.
Tasks stop shortly after starting	Container command, image, application startup, memory	Inspect stopped reason, exit code, and CloudWatch logs.
No targets appear	Service load-balancer integration or task registration	Confirm target group is attached to the ECS service and the selected container/port matches the task definition.
Port 80 targets appear unexpectedly	ASG direct target-group association	Check ASG Integrations and remove unintended direct registration.
Targets time out	Security groups, NACLs, route/network path, app listener	Allow ALB SG to reach the task host port; check app startup and target port.

Observed result	Likely area	Next action
Targets return 404	Wrong context path or health-check path	Test /hiring/ and configure target-group health checks to use the working endpoint.
Only some targets healthy	Task-specific startup, placement, resource pressure, or intermittent app errors	Compare each target's health reason, task logs, and instance resources.
ALB DNS fails to load	ALB state, listener, public subnets/routes, SG, DNS/URL typo	Confirm ALB is Active, listener :80 exists, subnets have appropriate internet routes, and inbound HTTP is allowed.

8. Operational notes and limitations

- The ECS service maintains the desired count by attempting to replace failed tasks; temporary shortfalls can occur.
- For genuine high availability, confirm that healthy targets exist in multiple AZs and that the Auto Scaling group can add capacity when needed.
- Dynamic host ports reduce fixed-port conflicts; they do not remove the need for correct security groups, health checks, and target registration.
- Using HTTP on port 80 is suitable for a simple lab. Production workloads should normally use HTTPS, certificates, restricted access, secure image management, and appropriate logging/monitoring.
- An inbound All TCP rule sourced from the ALB security group is broad. Restrict it to the required dynamic port range where practical and compatible with the ECS configuration.
- The phrase “Free Tier” depends on account age, region, current AWS offers, and resource type. Check the Billing console and pricing pages before leaving EC2, ALB, storage, public IPv4, or logging resources running.
- Do not attach the same target group to the Auto Scaling group for direct instance registration when the ECS service is expected to register task targets dynamically.
- A WAR filename commonly maps to a Tomcat context path, but confirm the actual application endpoint because the image could customize Tomcat deployment behavior.

9. Final result and conclusion

According to the supplied lab validation, the final state reached was six ECS tasks running, six target-group targets healthy, and the Hiring application accessible through the ALB DNS name at /hiring/. The deployment demonstrates the combined use of ECS on EC2, an Application Load Balancer, multi-AZ placement, and dynamic host-port mapping.

The five key lessons are:

7. Match the ECS container port to the port on which the application actually listens (8080 in this lab).
8. Use hostPort 0 with bridge networking for dynamically assigned host ports.
9. Allow the ALB security group to reach the registered host ports on the EC2 instances.
10. Let ECS manage task target registration; avoid unintended direct ASG registration to the same target group.
11. Set the health-check path to a valid application endpoint (/hiring/ for this lab), not automatically to /.

Replication note
This runbook reflects the configuration and outcome supplied in the request. AWS console labels and account-specific behavior can vary, and no live access to the account was performed while preparing this document. Validate each resource in the target account before treating the result as reproduced.

Appendix A — Useful commands

Run commands from an EC2 container instance only when you have authorized shell access. A dynamically allocated port is an example, not a permanent value.

```
# Check the application root path (a 404 can indicate the wrong context path)
curl -v http://localhost:32768/

# Check the expected Tomcat application context path
curl -v http://localhost:32768/hiring/

# Inspect task and container logs in the AWS console
CloudWatch → Log groups → /ecs/hiring-app-task
```

Replace 32768 with the actual host port shown for the target/task on the instance. A 404 on / alone does not prove the app is healthy; test /hiring/ and inspect the HTTP status. A successful response on the context path supports using it for the target-group health check.

Appendix B — Screenshot evidence checklist

#	Screenshot to capture	What it should prove
1	EC2 → Target Groups → hiring-app-tg → Details	Target type Instances; target group protocol/port details.
2	EC2 → Load Balancers → hiring-app-alb	ALB Active, internet-facing, subnets/AZs, DNS name.
3	EC2 → Security Groups → hiring-app-alb-sg	Inbound HTTP :80 rule and attached ALB SG.
4	EC2 → Instances	Running ECS container instances, instance type, and instance SG.
5	ECS → Cluster → Infrastructure / container instances	Instances registered with hiring-app-cluster.
6	ECS → Task definitions → hiring-app-task → JSON	bridge network mode, containerPort 8080, hostPort 0.
7	ECS → Cluster → Services → hiring-app-service	Desired count 6, running count 6, load-balancer integration.
8	ECS → Tasks / task details	Task placement across AZs and container instances.
9	EC2 → Target Groups → Targets	Six registered targets, dynamic ports, all Healthy.
10	EC2 → Target Groups → Health checks	Path /hiring/ and success code 200.
11	CloudWatch → Log groups → /ecs/hiring-app-task	Tomcat starts and listens on port 8080.
12	Browser showing ALB DNS + /hiring/	End-to-end application access through the ALB.